

Know When to *Switch*

Audit your own pipeline's memory, merges, and runtimes so you catch the switching signal before production catches it at 2 a.m.

NAME

DATE

RUNTIME (MIN)

1 Run the Column Audit, Not df.info()

- Pick your **largest dataframe** and run `df.memory_usage(deep=True)` — never trust `df.info()`.
- Rank columns by GB. In the post, *two* high-cardinality string columns ate 9 of 11 GB.
- If any single dataframe crosses **~20% of RAM**, name the culprit columns and whether `category` dtype fixes it.

```
df.memory_usage(deep=True).sort_values(ascending=False) / 1e9
df[col] = df[col].astype("category")
```

TOP OFFENDING COLUMNS

2 Hunt the Silent Merge That's Already Wrong

- Check **both sides** of one recurring merge with `df.dtypes` before it runs.
- Look for `int64` matched against `float64` or `object` — the post's merge silently dropped **8%** of rows for 11 days, no error.
- Row-count *before* and *after* the join. Write the delta, even if it's zero.

ROWS DROPPED, %

3 Race Pandas Against Polars on Your Own Job

- Take one **groupby-heavy job** and rewrite just the aggregation in Polars' lazy API.
- Time both versions on the *same input* — runtime and peak memory, side by side.
- The post went **52 min / 58 GB** → **4:10 / 11 GB**. Log your own numbers, even a small win counts.

```
pl.scan_parquet("file.parquet").group_by(cols).agg(...).collect()
```

BEFORE → AFTER

4 Match Your Job to Its Switching Signal

- Read the four signals: pandas (`memory_usage` past ~20% RAM), Polars (`scan_parquet` still can't finish, or repeated `.to_pandas()`), DuckDB (lock contention from concurrent writers), PySpark (JVM + shuffle time beats actual compute).
- Pick **one job you run regularly** — name the signal it's closest to hitting right now.
- Write the *next tool* it should move to once that signal fires.

JOB → NEXT TOOL

5 Start the One-Line Runtime Log

- For one recurring job, append **one line per run**: timestamp, runtime, input row count.
- After 5–10 runs, plot runtime against row count.
- Still a *straight* line, or has it started bending? That bend is the exact cue to switch — before it becomes a **2 a.m.** page.

```
2026-07-29 09:14, 41s, 2_140_000 rows
```

STRAIGHT OR BENDING?